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MAE 5730
Intermediate Dynamics and Vibrations
Problem 16
Due 10/14/17

MAE 5730

Intermediate Dynamics and Vibrations

Problem 16

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Problem 16: Two Masses

This problem has 3 independent educational goals:

(a) Motivate the use of kinematic constraints.

(b) Introduce the simplest of a class of vibrations problems which you should master. At this point, it is this aspect: mastery of derivation of the equations of motion. You should check that you can reproduce the lecture example with *no* sign errors without looking up anything.

(c) Development of critical exploration using analytical and numerical methods.

Two masses m_1 and m_2 are constrained to move frictionlessly on the x axis. Initially they are stationary at positions $x_1(0) = 0$ and $x_2(0) = \ell_0$. They are connected with a linear spring with constant k and rest length ℓ_0. A rightwards force is applied to the second mass. It is a step, or 'Heaviside function'

$$F(t) = F_0 H(t) = \begin{cases} 0 & \text{if } t < 0 \\ F_0 & \text{if } t \geq 0 \end{cases} \tag{1}$$

(a) Write code to calculate, plot and (optionally) animate the motions for arbitrary values of the given constants.

(b) Within numerical precision, should your numerical solution always have the property that $F = (m_1 + m_2)a_G$ where $x_G = (x_1 m_1 + x_2 m_2)/(m_1 + m_2)$? (As always in this course, "yes or no" questions are not multiple choice, but need justification that another student, one who got the opposite answer as you, would find convincing.)

(c) Use your numerics to demonstrate that if k is large the motion (displacement relative to its starting position) of *each* mass is, for time scales large compared to the oscillations, close to the center of mass motion.

(d) 5730 only: Using analytic arguments, perhaps inspired by and buttressed with numerical examples, make the following statement as precise as possible:

For high values of k the system nearly behaves like a single mass.

Of course, in detail, the system has 2 degrees of freedom (DOF). So you are looking for a way to measure the extent to which the system acts like it has 1 DOF, and in which conditions (for which extreme values of parameters and times) the system is close to 1 DOF by that measure. There is not a simple single unique answer to this question.

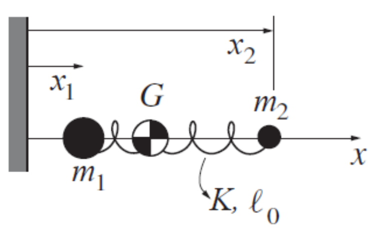

Figure 1: Two masses connected by a stiff spring in 1D.

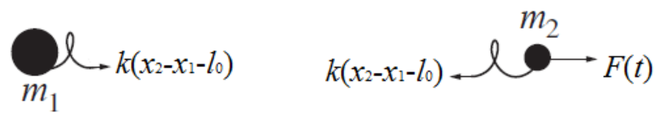

Figure 2: Free-body diagrams of the two masses and the forces acting upon them.

Part a

Refer to Fig. 2 for free-body diagrams that allows the linear momentum balance equations to be written:

$$m_1\ddot{x}_1 = k\big[(x_2 - x_1) - \ell_0\big] \tag{2a}$$
$$m_2\ddot{x}_2 = F(t) - k\big[(x_2 - x_1) - \ell_0\big] \tag{2b}$$

Defining $z_1 \equiv x_1$, and $z_2 \equiv x_2$, this is written in the following form suitable for a numerical solver:

$$\boxed{\begin{aligned}
\dot{z}_1 &= z_3 && \text{(3a)}\\
\dot{z}_2 &= z_4 && \text{(3b)}\\
\dot{z}_3 &= \frac{k}{m_1}\big[(z_2 - z_1) - \ell_0\big] && \text{(3c)}\\
\dot{z}_4 &= \frac{F(t)}{m_2} - \frac{k}{m_2}\big[(z_2 - z_1) - \ell_0\big] && \text{(3d)}
\end{aligned}}$$

Fig. 3 plots the position of each mass and their center of mass, given by

$$x_G = \frac{m_1x_1 + m_2x_2}{m_1 + m_2} \tag{4}$$

as a function of time. Fig. 4 plots the difference between each pair of the positions as a function of time.

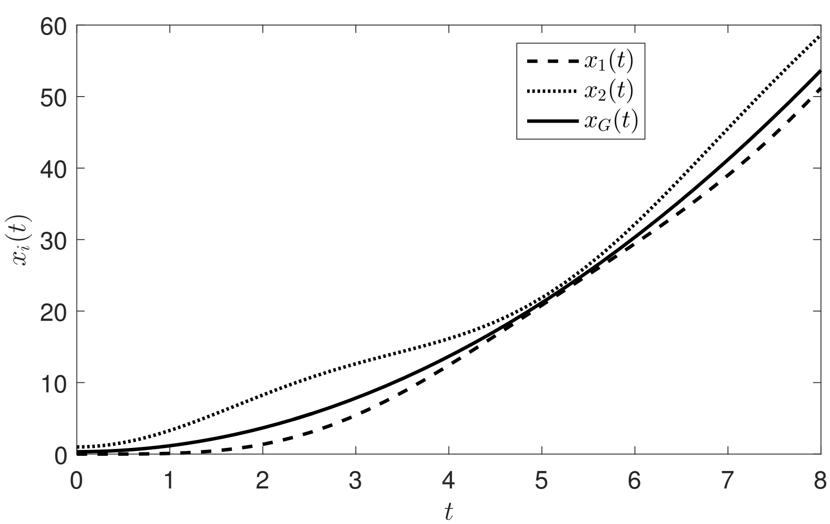

Figure 3: Plot of the position of each mass and their center of mass as a function of time for $m_1 = 2$, $m_2 = 1$, $F_0 = 5$, $\ell_0 = 1$, and $k = 1$. The three positions become close at times and far apart at other times, which is better visualized in a plot of the difference in their positions, shown by Fig. 4.

Below is the main `MATLAB` script `p16.m` used to accomplish Part a and Part c of this problem:

```
% Michael R. Buche
```

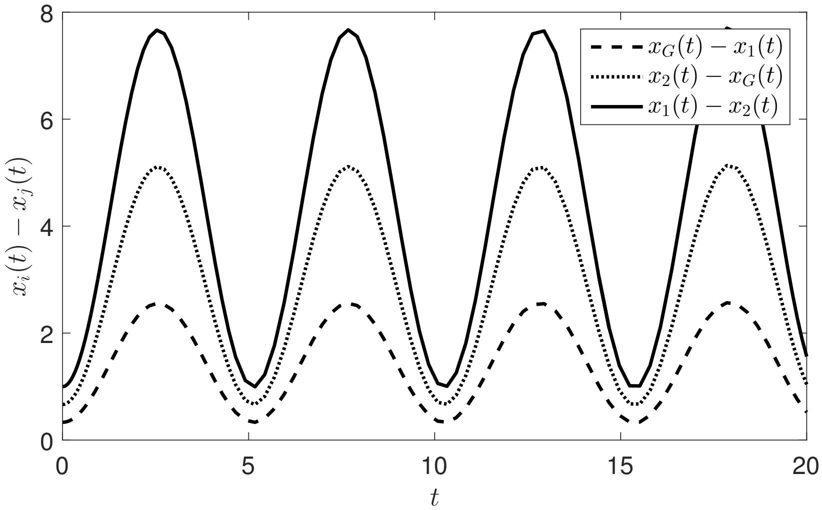

Figure 4: Plot of the difference in position between each pair of the two masses and the center of mass as a function of time for $m_1 = 2$, $m_2 = 1$, $F_0 = 5$, $\ell_0 = 1$, and $k = 1$. The difference sinusoidally varies as the two masses vibrate via the spring between them.

```
% MAE 5730 - Problem 16
% Two Masses

clear; close all; clc;

% System parameters
p.m1 = 2;
p.m2 = 1;
p.F0 = 5;
p.l0 = 1;
p.k = 1;

% Initial conditions and timespan
x0 = [0 p.l0];
v0 = [0 0];
icv = [x0 v0];
tspan = [0 8];

% ODE45 accuracy settings
opts.reltol = 1e-6;
opts.abstol = 1e-6;

% Use the ODE45 solver with the right hand side
[t,z] = ode45(@p16_RHS,tspan,icv,opts,p);
x1 = z(:,1);
x2 = z(:,2);
xG = (x1*p.m1 + x2*p.m2)/(p.m1+p.m2);

% Plot the motions
fontsize = 14;
figure
plot(t,x1,'k--',t,x2,'k:',t,xG,'k','linewidth',2)
```

```
legendo = legend('$x_1(t)$','$x_2(t)$','$x_G(t)$');
set(legendo,'Interpreter','latex','FontSize',fontsize)
xlabel('$t$','Interpreter','LaTeX');
ylabel('$x_i(t)$','Interpreter','LaTeX');
set(gca,'fontsize',fontsize);

% Plot position differences
figure
plot(t,xG-x1,'k--',t,x2-xG,'k:',t,x2-x1,'k','linewidth',2)
legendo = legend('$x_G(t)-x_1(t)$','$x_2(t)-x_G(t)$','$x_1(t)-x_2(t)$');
set(legendo,'Interpreter','latex','FontSize',fontsize)
xlabel('$t$','Interpreter','LaTeX');
ylabel('$x_i(t)-x_j(t)$','Interpreter','LaTeX');
set(gca,'fontsize',fontsize);
```

Below is the secondary `MATLAB` function `p16_RHS` that is called by the above main script to solve Eqs. (3):

```
% Michael R. Buche
% MAE 5730 - Problem 16
% Two Masses
% Right hand side to be called

function xprime = p16_RHS(t,z,p)
    dx1 = z(3);
    dx2 = z(4);
    ddx1 = p.k*(z(2)-z(1)-p.l0)/p.m1;
    ddx2 = (p.F0*heaviside(t) - p.k*(z(2)-z(1)-p.l0))/p.m2;
    xprime = [dx1; dx2; ddx1; ddx2;];
end
```

Part b

Add Eqs. (2) together, where the spring force terms are easily seen to cancel. The result is

$$m_1\ddot{x}_1 + m_2\ddot{x}_2 = F(t) \tag{5}$$

Rewrite the left-hand side using $a_G = \ddot{x}_G = \frac{m_1\ddot{x}_1+m_2\ddot{x}_2}{m_1+m_2}$

$$\boxed{(m_1+m_2)\ddot{x}_G = (m_1+m_2)a_G = F(t)} \tag{6}$$

This is independent of any parameter choice, so the numerical solution should always have this property as well within numerical precision.

Part c

See Fig. 5 for a plot of the motions for $m_1 = 2$, $m_2 = 1$, $F_0 = 5$, $\ell_0 = 1$, and $k = 1000$. The plots show that the distance between the masses and their center of mass remains constant neglecting the small oscillations seen on the right at very small time scales. If their relative positions remains constant in time, their time derivatives (velocity and acceleration) are equal, and therefore the motion of each match matches that of their center of mass.

Part d

Divide Eqs. (2) by k and define the small perturbation parameter $\epsilon^2 \equiv m_1/k$ and mass ratio $\alpha \equiv m_2/m_1$ in order to receive

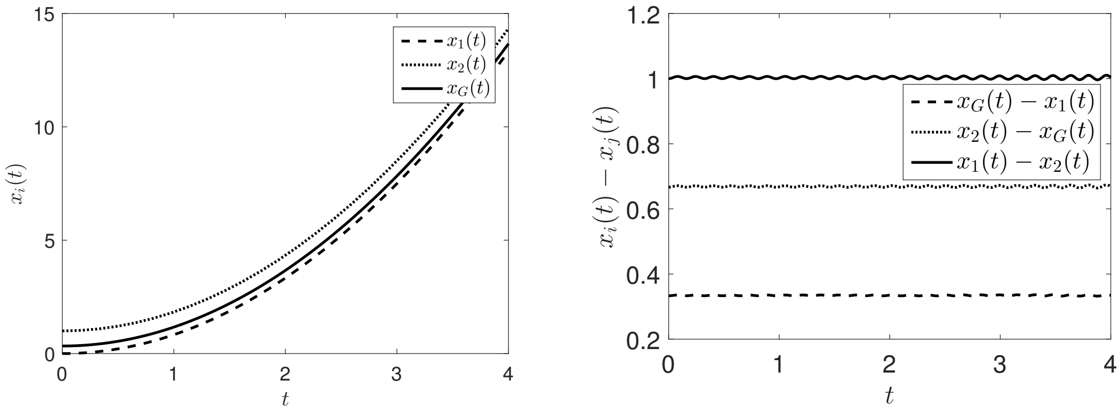

Figure 5: Plot of the position of each mass and their center of mass (left) and of the difference in position between each pair of the two masses and the center of mass (right) function of time for $m_1 = 2$, $m_2 = 1$, $F_0 = 5$, $\ell_0 = 1$, and $k = 1000$. The distance between the masses and between them and their center of mass remains constant in time, neglecting the small oscillations seen on the right at very small time scales. This shows that the motion of each mass matches that of their center of mass, since their velocity and acceleration must match if their relative positions remains constant in time.

$$\epsilon^2 \ddot{x}_1 = (x_2 - x_1) - \ell_0 \tag{7a}$$

$$\alpha\epsilon^2 \ddot{x}_2 = \frac{F_0}{k}\, H(t) - (x_2 - x_1) + \ell_0 \tag{7b}$$

This is a singular perturbation problem since ϵ multiplies the highest-order term in both equations. The leading order behavior of perturbation problems (in regular perturbation problems, found by solving with the perturbation parameters going to zero, or equivalently expanding the dependent variables in powers of the perturbation parameters and solving at each power) exemplifies the system behavior when the parameters are approximately zero. Taking $\epsilon = 0$ and $F_0/k = 0$ uncovers the following leading order behavior in both equations:

$$\boxed{x_2 - x_1 = \ell_0} \tag{8}$$

This seems to show that to leading order, which is high enough values of k such that m_1/k, m_2/k, and F_0/k are all extremely small, the system behaves like a single mass. Since this is a singular perturbation problem though, a great loss of information about the system is experienced for $\epsilon = 0$. For example, consider the algebraic equation

$$\epsilon x^2 + x + 1 = 0 \tag{9}$$

where $\epsilon \ll 1$. This equation has two roots, but solving the leading order equation via $\epsilon = 0$ only yields one unperturbed root, $x = -1$. Any information about the other root has been lost, and in fact, it blows up as $-\epsilon^{-1}$ when $\epsilon \to 0$. Because of this phenomenon, it was a mistake to write Eq. (8) in this fashion, although in this case it happens to have the correct behavior. Rescaling will allow the equation to be analyzed more fully, also allowing a more appropriate comparison of small terms to terms on the order of unity. Divide Eqs. (7) by ℓ_0 and define the non-dimensional positions $X_i \equiv x_i/\ell_0$ in order to write

$$\epsilon^2 \ddot{X}_1 = (X_2 - X_1) - 1 \tag{10a}$$

$$\alpha\epsilon^2 \ddot{X}_2 = \frac{F_0}{k\ell_0}\, H(t) - (X_2 - X_1) + 1 \tag{10b}$$

Further rescaling must be done: define the non-dimensional time $\tau \equiv t/\epsilon$ and the non-dimensional force small perturbation parameter $\epsilon_F \equiv F_0/k\ell_0$ in order to write the following:

$$\frac{\partial^2 X_1}{\partial \tau^2} = (X_2 - X_1) - 1 \tag{11a}$$

$$\alpha \, \frac{\partial^2 X_2}{\partial \tau^2} = \epsilon_F \, H(\tau) - (X_2 - X_1) + 1 \tag{11b}$$

These are non-dimensional equations, and pose a regular perturbation problem in ϵ_F. The leading order behavior is discovered when $\epsilon_F = 0$ (appropriately taken here), which is to say $F_0 \ll k$, which yields

$$\frac{\partial^2 X_1}{\partial \tau_1^2} = (X_2 - X_1) - 1 \tag{12a}$$

$$\alpha \, \frac{\partial^2 X_2}{\partial \tau_2^2} = 1 - (X_2 - X_1) \tag{12b}$$

The problem can be finished by considering the order of magnitude of the non-dimensional terms above. The non-dimensional positions X_i are on the order unity since $X_i \equiv x_i \ell_0$ was taken and ℓ_0 was the only relevant length scale to start. The non-dimensional time $\tau^2 = t^2/\epsilon = t^2 k/m_1$ is much greater than order unity for $t = \text{ord}(1)$ when for $m_2/m_1 = $ord(1), $k \gg m_1$ and becomes infinitely large instead. Then $X_i = \text{ord}(1)$ and $\tau^2 \gg \text{ord}(1)$ causes the derivatives on the left-hand sides to be negligible in comparison to the right-hand sides, which allows one to obtain $X_2 - X_1 = 1$ twice, which means the leading order behavior is again

$$\boxed{x_2 - x_1 = \ell_0} \tag{13}$$

This is only valid for large time scales though, and does not capture the system behavior for small time scales. This is understood by examining Fig. 5, where there is oscillation on small time scales that is not seen at large time scales. This is the information that is lost when taking $\epsilon = 0$ in the original singular perturbation problem. Differentiating shows $\dot{x}_1 = \dot{x}_2$ and $\ddot{x}_1 = \ddot{x}_2$, which in turn shows that the center of mass motion is also equal:

$$\boxed{\dot{x}_G = \frac{m_1 \dot{x}_1 + m_2 \dot{x}_2}{m_1 + m_2} = \frac{(m_1 + m_2)\dot{x}_1}{m_1 + m_2} = \dot{x}_1 = \dot{x}_2 \qquad \text{and} \qquad \ddot{x}_G = \ddot{x}_1 = \ddot{x}_2} \tag{14}$$

It has been shown that the leading order behavior, which is high enough values of k such that m_1/k, m_2/k, and $F_0/k\ell_0$ are all extremely small compared to 1, the system behaves like a single mass. This has been accomplished by reducing the equations to order unity through non-dimensionalization and analyzing their leading order behaviors using perturbation methods. To measure the extent to which the system acts like it has 1 DOF and behaves as a single mass, one can compute $\epsilon = m_1/k$, $\alpha = m_2/m_1$, and $\epsilon_F = F_0/k\ell_0$ and verify that $\epsilon \ll 1$, $\alpha = \text{ord}(1)$, and $\epsilon_F \ll 1$ and equate the measure to how small the ϵ's actually are.